MATH 33A: Linear Algebra and Applications

Fall Quarter – Sample Syllabus (fictional, for testing)

Instructor: Dr. A. Example **Office hours:** Tuesdays 2–4pm

Grading

Homework 20%, Midterm 1 20%, Midterm 2 20%, Final 40%. Late homework is not accepted.

Course Schedule

- Week 1: Vectors in R^n, vector addition and scalar multiplication, linear combinations and span.
- Week 2: Systems of linear equations, row reduction, echelon forms.
- Week 3: Matrix-vector products; linear transformations of the plane (rotations, shears, reflections).
- Week 4: Matrix multiplication as composition of transformations. Midterm 1.
- Week 5: Inverse matrices; the determinant as an area scaling factor.
- Week 6: Subspaces, basis, dimension; column space and null space.
- Week 7: Eigenvalues and eigenvectors; characteristic polynomial. Midterm 2.
- Week 8: Diagonalization and change of basis.
- Week 9: Dot product, orthogonality, orthogonal projections, Gram–Schmidt.
- Week 10: Least squares and linear regression. Review.